

	temperature of a substance is directly proportional to the average kinetic energy of the molecules in the substance. As the temperature is the same for both ice and water, the average kinetic energy of their molecules would be the same.	B1 B1
7a	Temperature of a substance <u>is directly proportional to the average kinetic energy</u> of the molecules in the substance. As the <u>temperature is the same</u> for both ice and water, the <u>average kinetic energy of their molecules would be the same</u>.	B1
7bi	Setup B helps to measure amount of melted ice due to <u>heat gained by the ice from the surroundings</u> .	A1
7bii	Thermal Energy from heater = Heat gained by ice $(12.0)(10 \times 60) = [(40.0 - 18.4) \times 10^{-3}] L_f$ $L_f = 3.33 \times 10^5 \text{ J kg}^{-1}$	M1 A1
7ci	$P = \frac{nRT}{V}$ $P = \frac{(0.200)(8.31)(27 + 273.15)}{5.00 \times 10^{-3}}$ $= 9.98 \times 10^4 \text{ Pa}$	A1
7cii	New Pressure $P_{\text{new}} = \frac{(0.200)(8.31)(27 + 273.15)}{(5.00 + 3.00) \times 10^{-3}}$ $= 6.2356 \times 10^4 \text{ Pa}$ $= 6.24 \times 10^4 \text{ Pa}$ $n_B = \frac{P_{\text{new}} V}{RT}$ $n_B = \frac{(6.2356 \times 10^4)(3.00 \times 10^{-3})}{(8.31)(27 + 273.15)}$ $= 0.0750 \text{ mol}$	A1 A1
7ciii	With the valve open, the molecules will move between the two containers till the pressure is equalized. $n_A + n_B = \frac{PV_A}{RT_A} + \frac{PV_B}{RT_B}$ $0.200 (8.31) = P \left(\frac{5.00 \times 10^{-3}}{27 + 273.15} + \frac{3.00 \times 10^{-3}}{100 + 273.15} \right)$ $1.662 = P (2.4698 \times 10^{-5})$ $P = 6.73 \times 10^4 \text{ Pa}$	M1 A1
7di	pV (= 170 J) evaluated correctly for 3 readings taken correctly from graph (2 marks for 3 readings, 1 marks for 2 readings) Conclude that since pV values are the same (or some slight variation) hence it is isothermal (or not isothermal)	M1, M1 A1

7dii	Process <u>must be carried out (infinitely) slowly</u> OR <u>container must be a good conductor of heat</u> so as to <u>allow time for heat to enter into / exit from the system in surroundings of constant temperature maintained by a water bath (or equivalent)</u>	B1 B1
7diii	At constant pressure, $V \propto T$: $\frac{1.7 \times 10^5}{T_A} = \frac{3.4 \times 10^5}{385}$ $T_A = 193 \text{ K}$	M1 A1
7div	$w = -1.7 \times 10^5 \times 5 \times 10^{-4} = -85 \text{ J}$ $q + w = \Delta U$, therefore $\Delta U = 213 - 85$ $= 128 \text{ J}$ Students who use the method below to award max 2 marks Increase in internal energy of the gas $= \frac{3}{2} \times 1.7 \times 10^5 \times (5 \times 10^{-4})$ [M1] $= 127.5 \text{ J}$ [A1]	M1 M1 A1